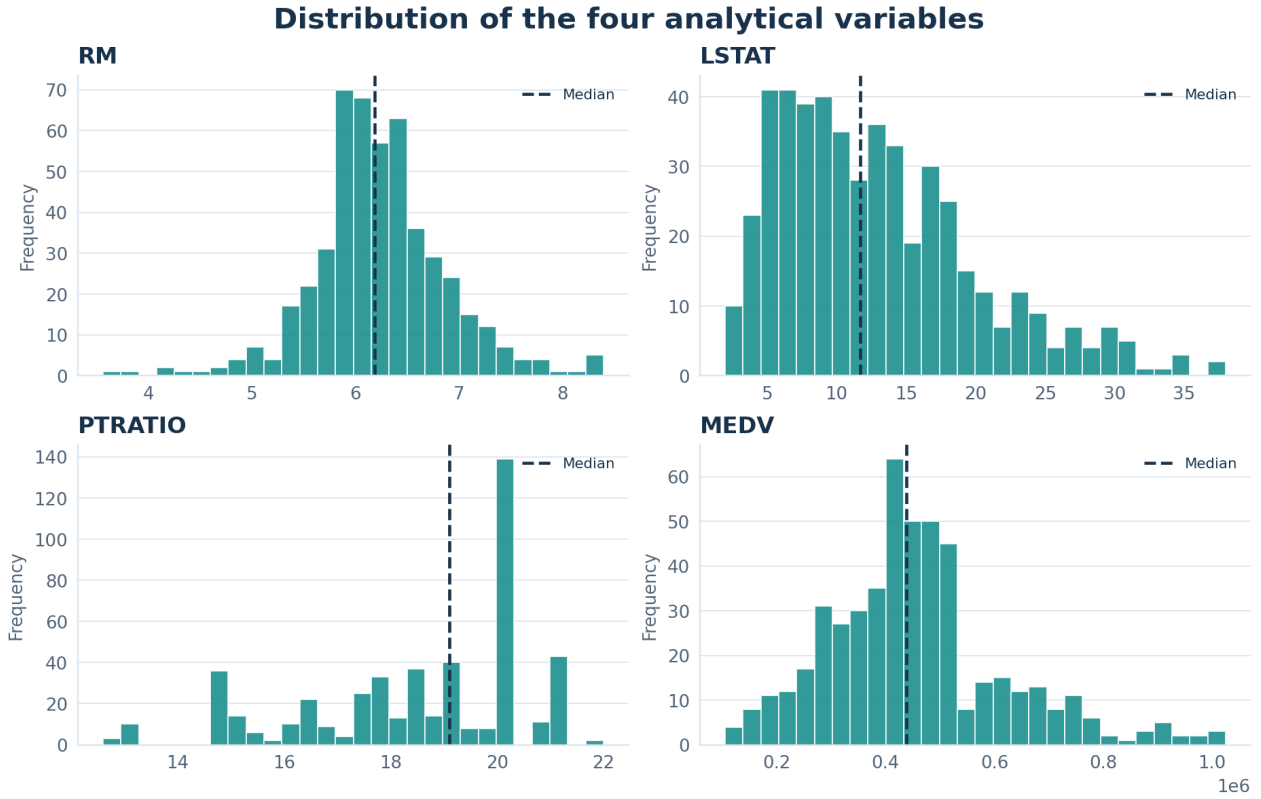
Exploratory Analysis & Predictive Modelling

A data-led explanatory report generated from the supplied Python dashboard and housing CSV dataset.

Executive overview

489 complete observations and four numeric fields are included. The dashboard uses RM, LSTAT and PTRATIO as explanatory variables and MEDV as the default target. The CSV contains **0 missing values** in these fields and **0 exact duplicate rows**.

Headline benchmark: Random Forest achieved the lowest holdout RMSE (R^2=0.851; MAE=44,053; RMSE=57,229) on the specified 80/20 split. This is a reproducible benchmark on one test partition, not a guarantee for new records.



Measurement caution: the code displays a dollar sign for MEDV in some dashboard labels, but the CSV does not document currency, geography, valuation date or measurement method. This report therefore treats MEDV as a recorded numeric target and does not assume a currency or location.

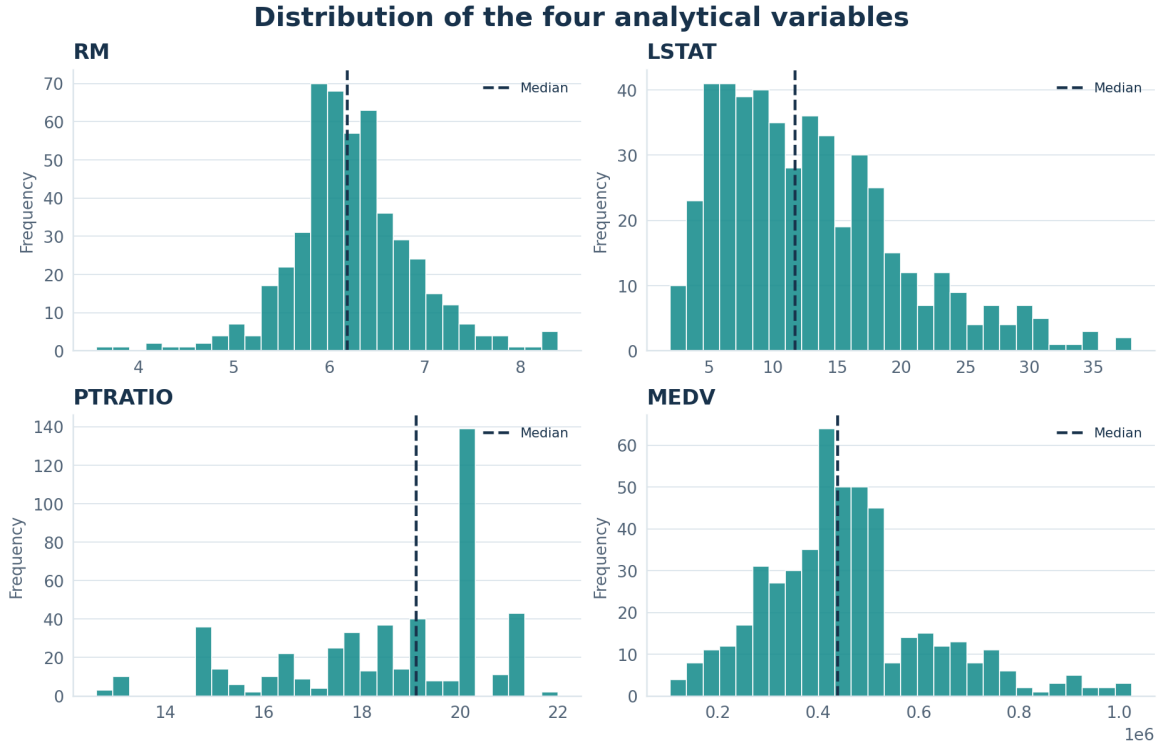
Variables and descriptive statistics

The variable descriptions below follow the definitions included in the supplied dashboard code.

What the fields represent

RM: average number of rooms per dwelling.
LSTAT: percentage indicator; the dashboard labels it as lower-status population percentage.
PTRATIO: pupil-teacher ratio.
MEDV: recorded median property-value target.

These four columns define the analytical scope. There are no geographic identifiers, property age, property type or documented currency/unit fields. Those attributes cannot be inferred from the supplied values.



Descriptive statistics

Variable	Count	Mean	Std. dev.	Minimum	Median	Maximum	Skewness
RM	489	6.24	0.64	3.56	6.18	8.40	0.08
LSTAT	489	12.94	7.08	1.98	11.69	37.97	0.91
PTRATIO	489	18.52	2.11	12.60	19.10	22.00	-0.82
MEDV	489	454,342.94	165,340.28	105,000.00	438,900.00	1,024,800.00	0.78

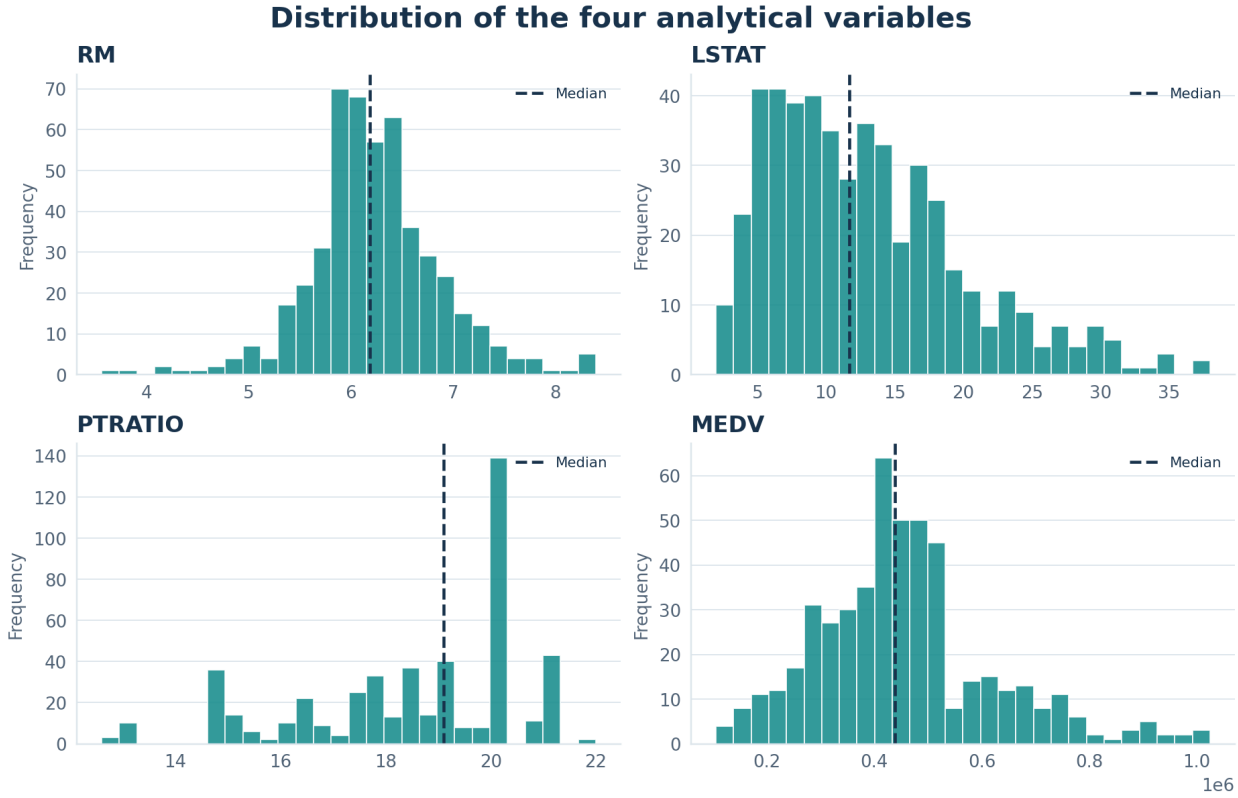
Distribution shape and potential outliers

Histograms show frequency patterns; box plots summarize medians, quartiles and unusual observations.

How to read these plots

Look for concentration, spread, skewness and possible multiple peaks. The box plot summarizes the median and interquartile range. The IQR screening rule flags observations outside Q1 – 1.5 × IQR or Q3 + 1.5 × IQR.

A flagged record is a candidate for review, not automatically an error. Confirm unusual values against source records before excluding them. Valid extreme observations may contain important information.



Potential outlier screening (1.5 × IQR)

Variable	Q1	Q3	Lower fence	Upper fence	Flagged
RM	5.88	6.58	4.84	7.62	22
LSTAT	7.37	17.12	-7.25	31.75	7
PTRATIO	17.40	20.20	13.20	24.40	13
MEDV	350,700.00	518,700.00	98,700.00	770,700.00	22

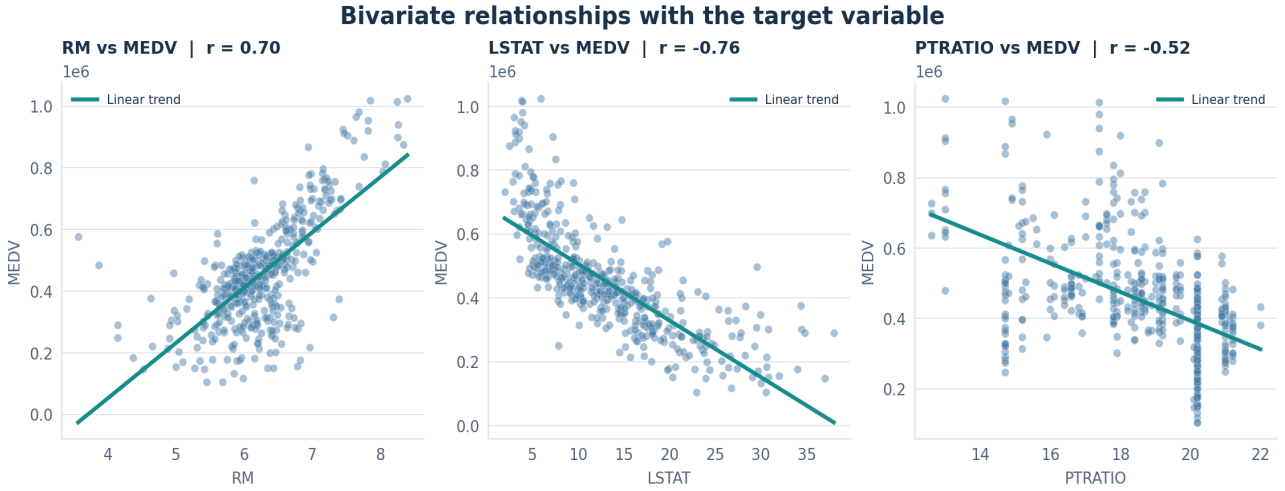
Relationships with the target variable

The scatter plots show observed records; trend lines summarize linear direction without implying causation.

Feature-by-feature interpretation

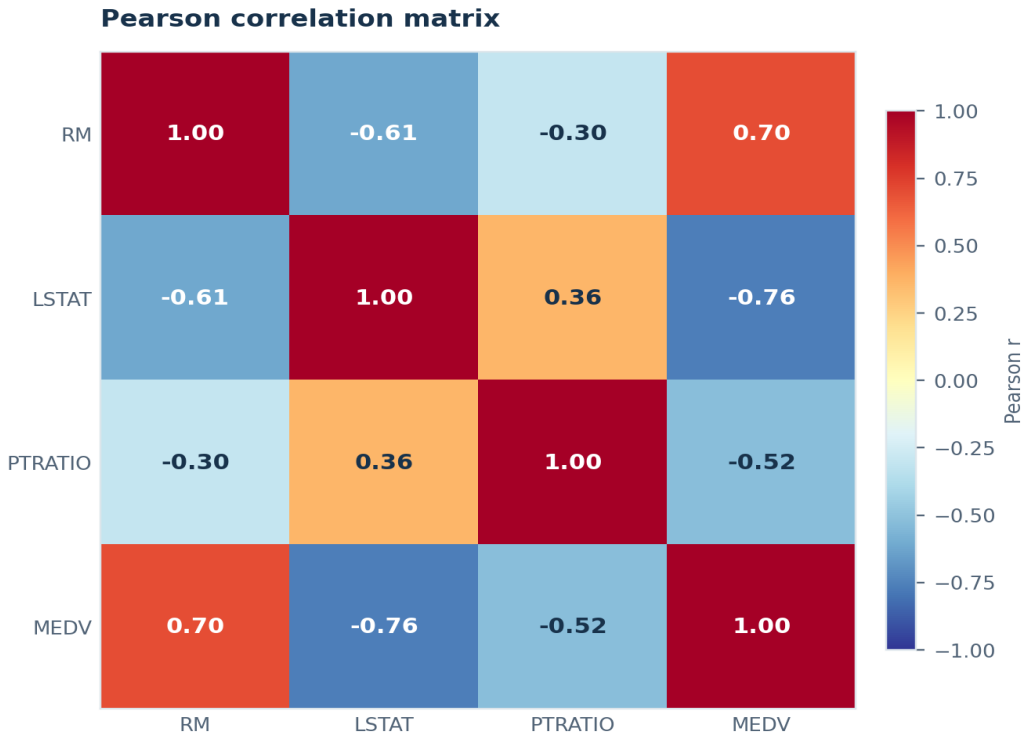
RM vs MEDV: positive, moderate Pearson association (r=0.697).
LSTAT vs MEDV: negative, strong Pearson association (r=-0.761).
PTRATIO vs MEDV: negative, moderate Pearson association (r=-0.519).

Pearson correlation measures linear association only. It does not prove that a feature causes a change in MEDV; nonlinear patterns may also exist.



Correlation matrix

The heatmap displays Pearson coefficients from –1 to +1. Values near +1 indicate positive linear association; values near –1 indicate negative association; values near zero indicate weak linear association. Use this matrix to inspect overlap among predictors and identify relationships worth investigating further.

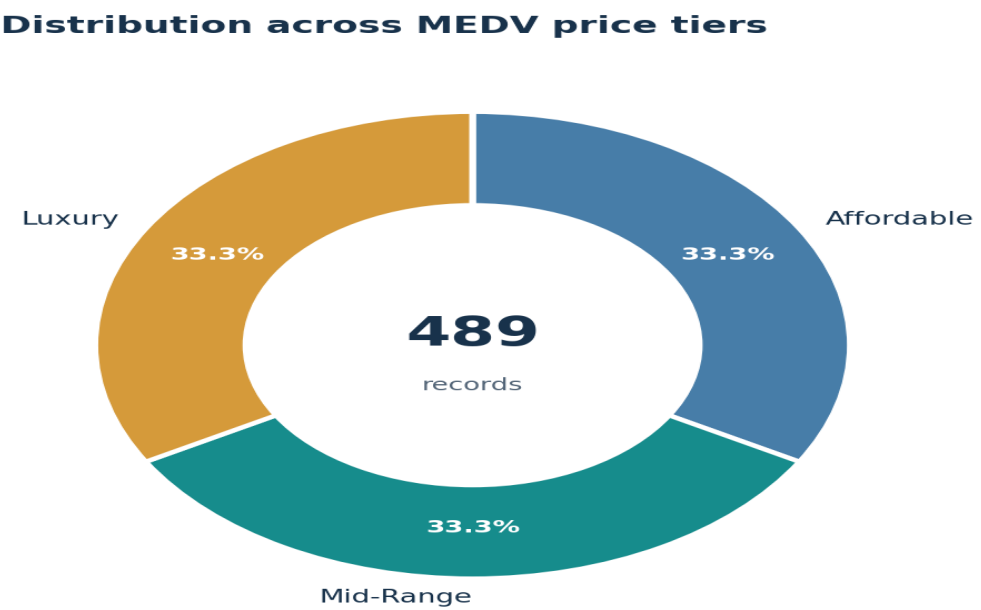


Relative price tiers and multivariable views

The dashboard derives three price groups from MEDV ranks; they are relative categories within this dataset.

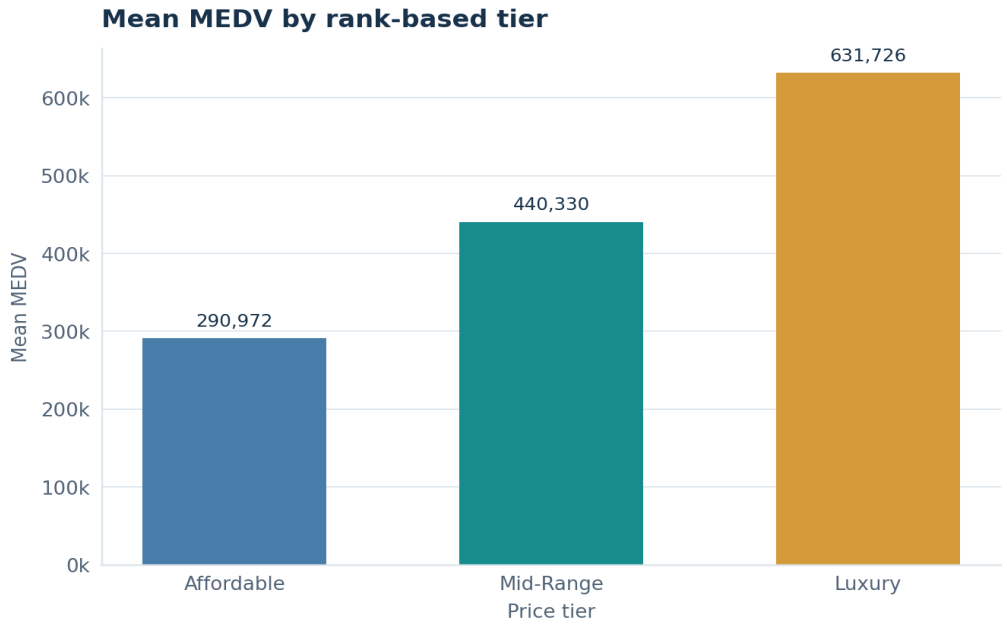
Tier distribution

Affordable, Mid-Range and Luxury tiers are created by dividing ranked MEDV observations into three approximately equal groups. They are not externally defined market segments. The donut chart shows the share of records assigned to each group. Because MEDV itself defines the groups, differences in the target across tiers are expected by construction.



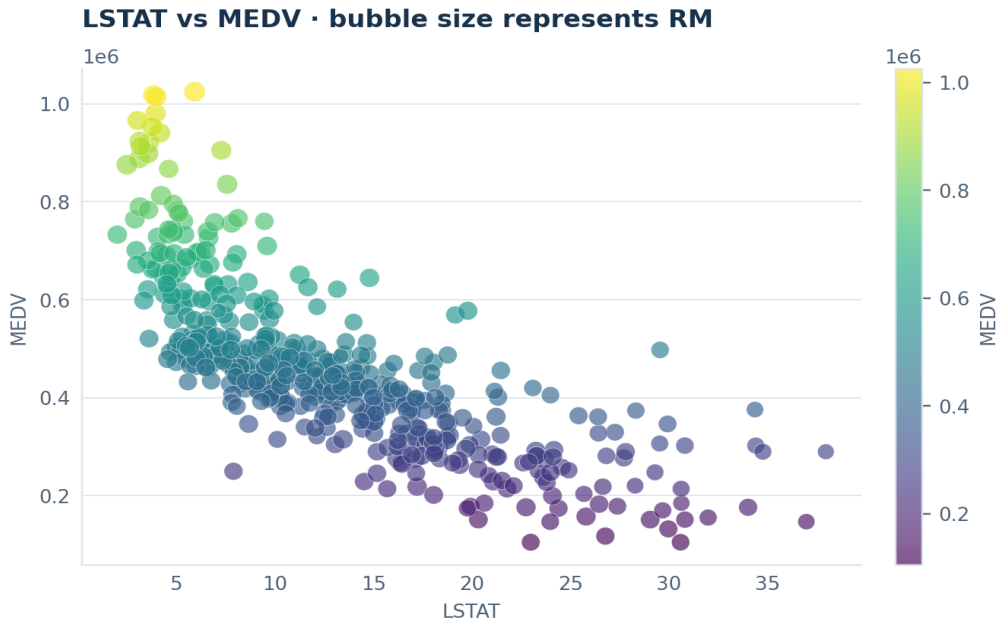
Mean MEDV by tier

This bar chart compares average MEDV across the rank-based groups. It is a descriptive summary, not independent validation of segmentation. Read it alongside the distributions to understand within-tier variability.



Three-variable bubble view

The bubble plot maps LSTAT to the horizontal axis and MEDV to the vertical axis; marker size represents RM and marker color represents MEDV. It adds a third feature, but overlap can hide individual records.



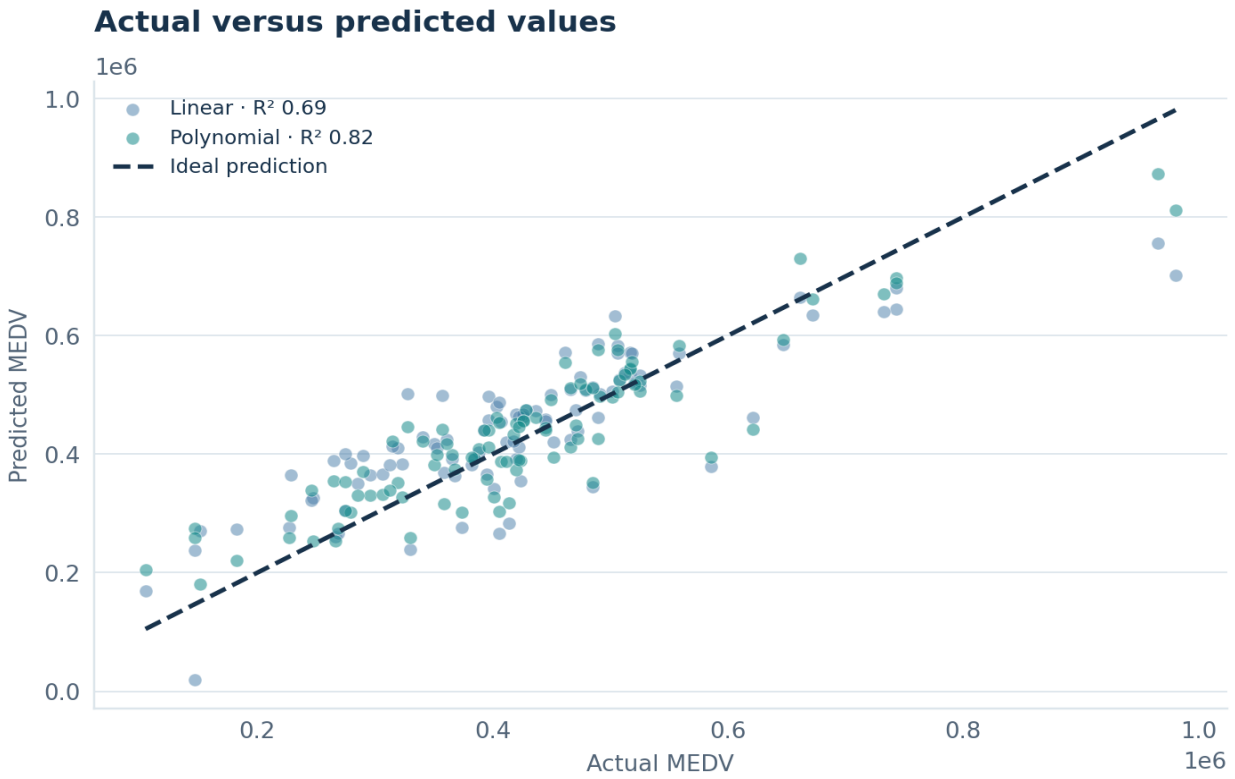
Linear and polynomial regression

Both models use the same 80/20 holdout split (random_state=42) and the three predictors RM, LSTAT and PTRATIO.

Actual versus predicted

Linear regression R^2: 0.691.
Polynomial degree-2 R^2: 0.818.

The polynomial pipeline expands the predictors with degree-two terms before fitting a linear estimator. The dashed diagonal represents perfect prediction; points closer to it have smaller errors. The polynomial model performs better on this particular holdout split, but should be checked through cross-validation and a separate final test set.



Evaluation metrics

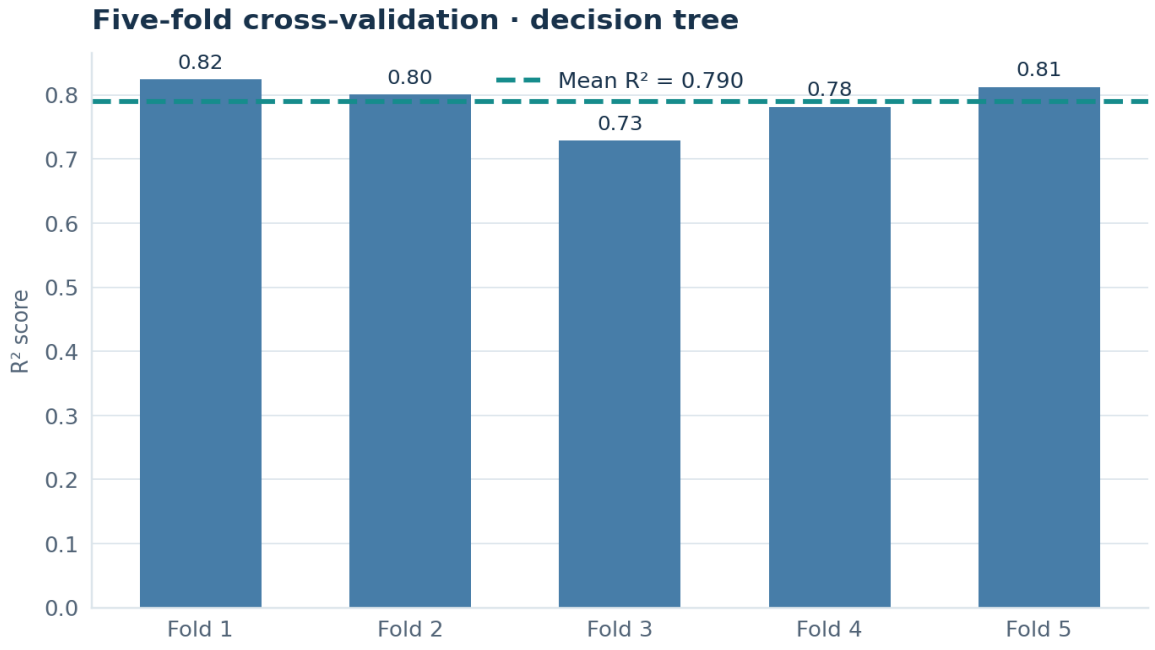
Metric	Meaning	Preferred direction
R^2	Proportion of target variance accounted for on the evaluation sample.	Higher is generally better; may be negative.
MAE	Average absolute prediction error, in MEDV units.	Lower is better.
RMSE	Square root of mean squared error; larger errors receive more weight.	Lower is better.

Cross-validation and model complexity

Validation estimates model stability and helps diagnose the balance between flexibility and generalization.

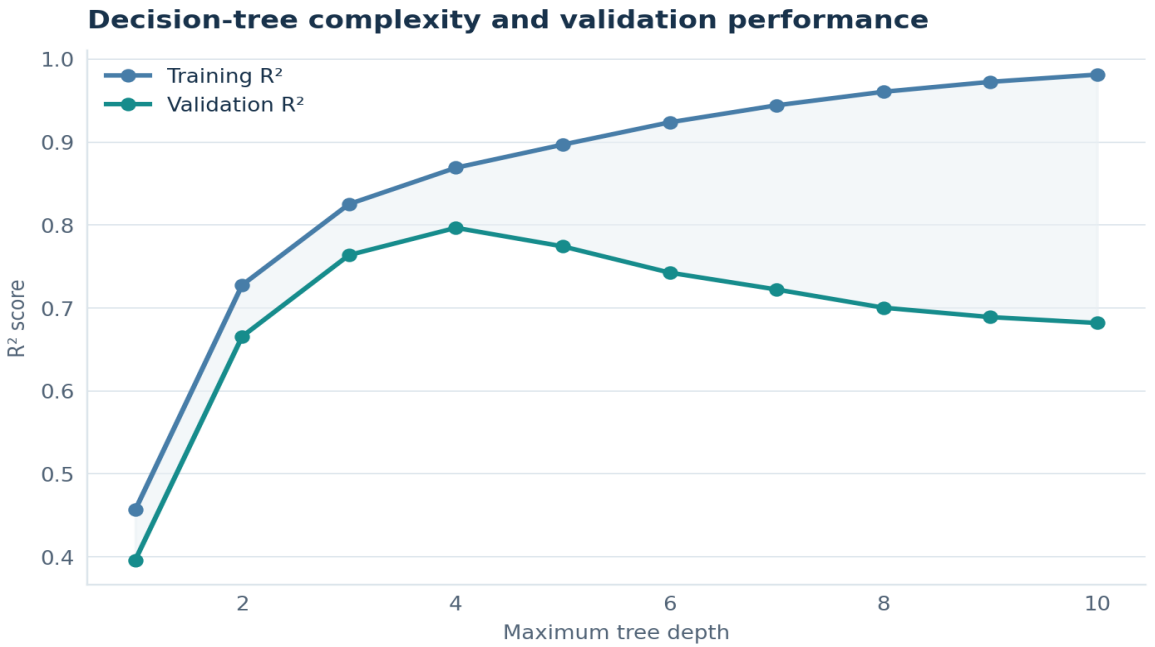
Five-fold cross-validation

The dashboard evaluates a depth-5 decision tree with shuffled five-fold K-Fold validation. Mean R^2 is **0.790** and fold-to-fold standard deviation is **0.033**. Each bar represents one validation fold. Similar scores suggest more stable performance across partitions; larger variation indicates sensitivity to the sampled folds.



Complexity and overfitting diagnostics

The validation curve changes maximum tree depth from 1 to 10. Training R² generally rises as the tree becomes more flexible. Validation R² may improve and then flatten or fall. A widening training-validation gap can indicate overfitting. The dashboard's 0.15 gap threshold is a practical heuristic, not a universal rule.



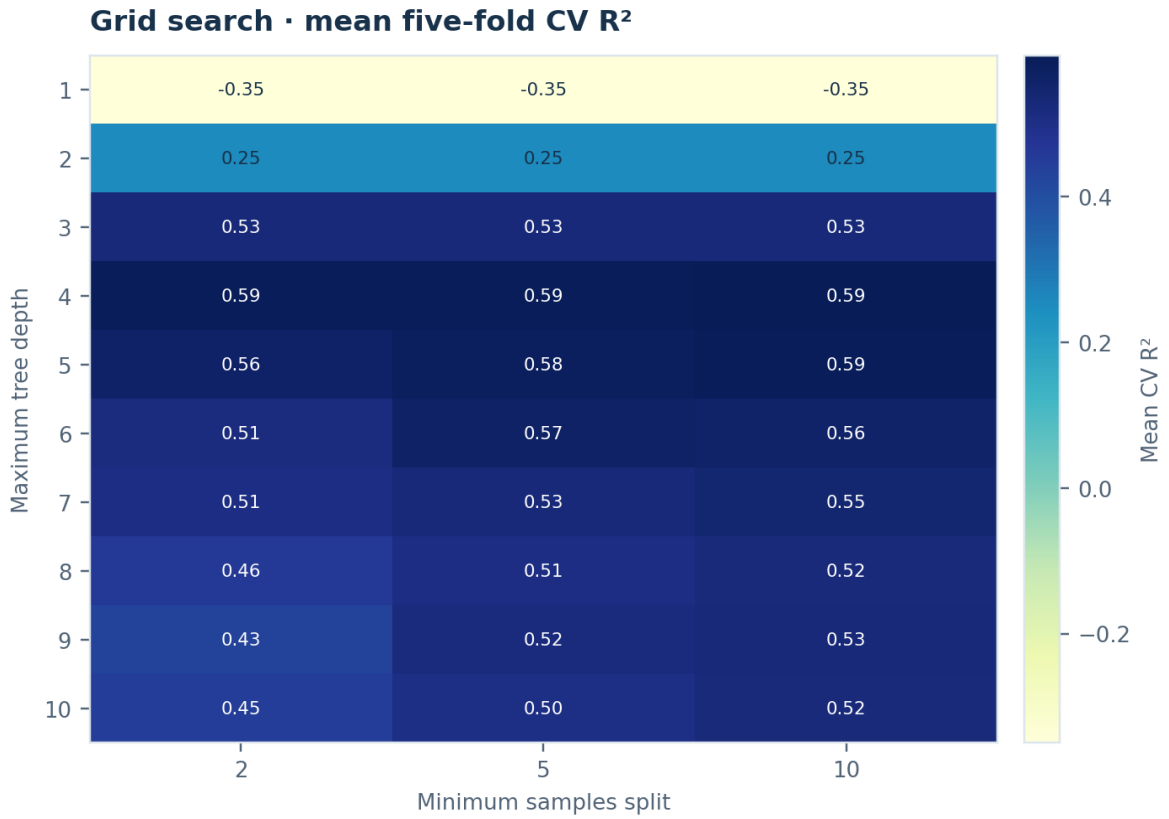
Decision-tree grid search

The heatmap reports mean five-fold cross-validation R^2 for the tested parameter combinations.

Selected parameter combination

The search tests maximum depths from 1 to 10 and minimum-sample-split values of 2, 5 and 10. Best combination: **max_depth=4** and **min_samples_split=10**, with mean CV R^2=0.594. A total of 30 combinations were evaluated.

This is the best result among the tested combinations, not necessarily a global optimum. The tuning score is not an unbiased final estimate; retain a separate untouched test set or use nested cross-validation.



How to interpret the heatmap

Compare cells using the shared color scale. Higher mean CV R^2 indicates better average performance for that parameter pair. If several settings are close, a simpler and shallower tree may be preferable for interpretability and stability.

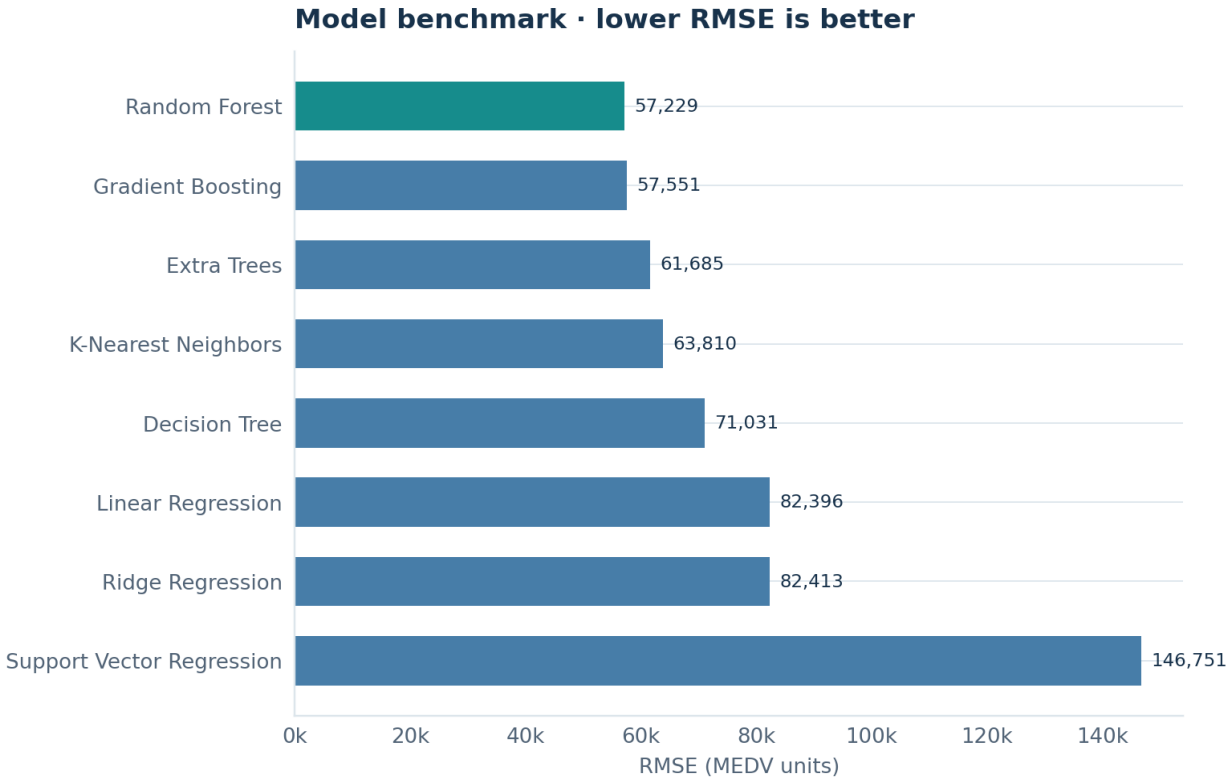
Eight regression algorithms on a common holdout set

All models use the same 80/20 train-test partition and random_state=42 for a fair comparison.

Benchmark interpretation

Top model by RMSE: Random Forest.
R²: 0.851
MAE: 44,053
RMSE: 57,229

Gradient Boosting is close to Random Forest in this run. The difference between the strongest and weakest algorithms shows that model choice matters for this dataset and split. The ranking may change with a different sample, new data or revised preprocessing.



Model performance table (sorted by RMSE)

Model	R²	MAE	RMSE
Random Forest	0.851	44,053	57,229
Gradient Boosting	0.849	44,113	57,551
Extra Trees	0.827	47,241	61,685
K-Nearest Neighbors	0.815	48,763	63,810

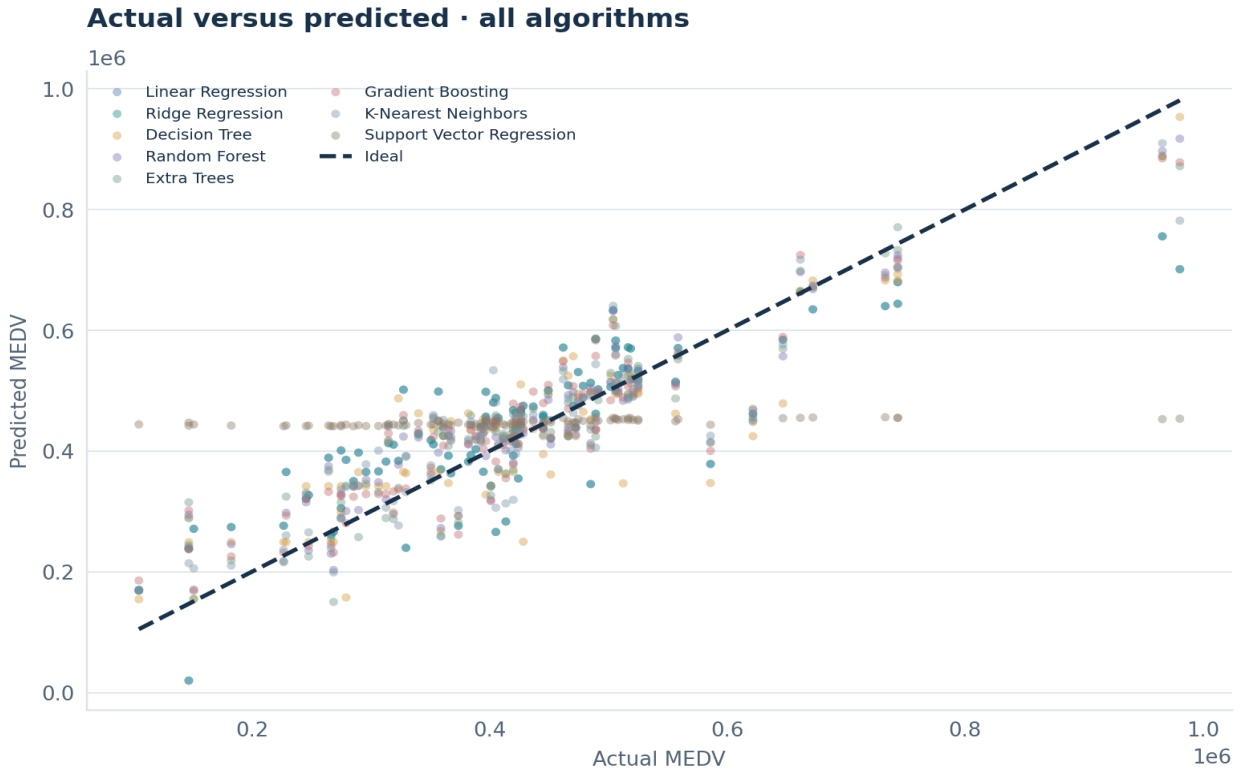
Model	R^2	MAE	RMSE
Decision Tree	0.770	53,041	71,031
Linear Regression	0.691	64,277	82,396
Ridge Regression	0.691	64,276	82,413
Support Vector Regression	0.020	106,379	146,751

Prediction fit and feature contribution

Use prediction plots to identify dispersion, systematic bias and model-specific feature contributions.

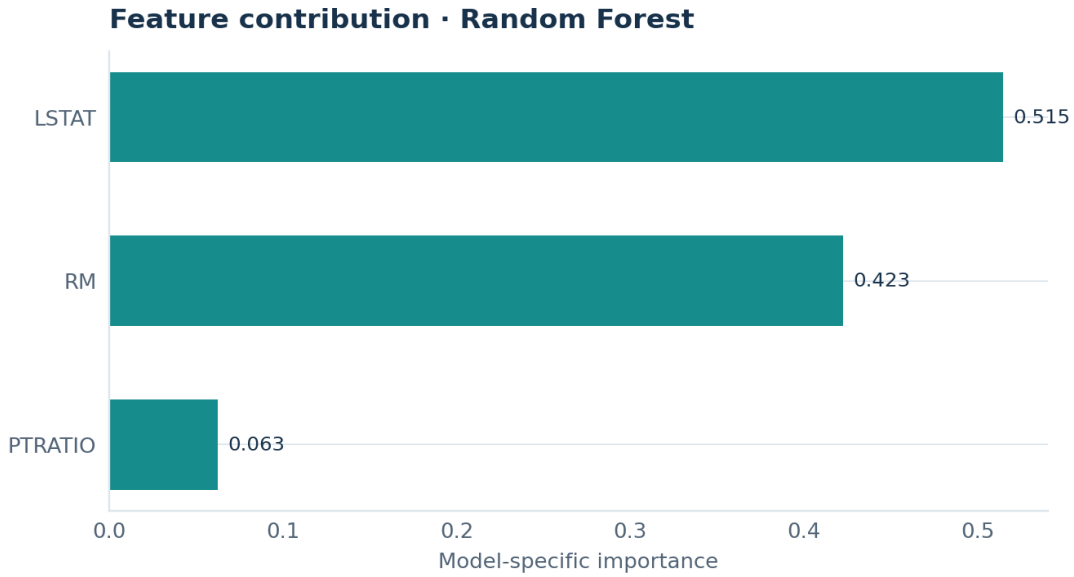
Actual versus predicted · all models

Each point compares a test-set actual value with a model prediction. The dashed diagonal represents perfect agreement. Points consistently above or below the line can indicate over- or under-prediction in parts of the target range. Because multiple algorithms overlap, inspect individual model plots before operational use.



Feature importance / absolute coefficients

The selected model is Random Forest. The chart reports model-specific feature importance or absolute coefficients, depending on estimator type. These measures are not necessarily comparable across algorithms and do not establish causal influence.



Evidence-based conclusions and next steps

Keep descriptive patterns, predictive performance and causal claims distinct.

Evidence	Interpretation
489 records; 4 numerical fields; no missing values or exact duplicates in required columns.	The analysis is complete for the four supplied variables, but the feature set is limited.
Random Forest achieved holdout R^2=0.851.	It ranked first by RMSE on this specific reproducible 80/20 split.
Polynomial R^2=0.818; linear R^2=0.691.	Degree-two terms improved holdout fit compared with the simple linear model in this run.
Grid-search best CV R^2=0.594.	The chosen settings were best among tested combinations, not necessarily globally optimal.
Correlation and feature importance are associational.	Neither proves that changing a predictor will cause MEDV to change.

Recommended workflow

1. **Document definitions:** confirm the meaning, unit, currency, date and provenance of MEDV and LSTAT.
2. **Review unusual records:** investigate IQR flags against source records; do not remove them automatically.
3. **Keep a baseline:** compare nonlinear algorithms with interpretable linear regression.
4. **Validate robustly:** use repeated or nested cross-validation and preserve a final untouched test set.
5. **Improve the feature set:** add variables only when definitions, quality and availability are documented. The current CSV has no location, age or property-type fields.
6. **Communicate uncertainty:** report errors in the documented target units and avoid presenting model estimates as formal property valuations.

Reproducibility: the supplied dashboard uses fixed random seeds for its principal split and model settings. This report was calculated from the supplied CSV using the principal algorithms and parameter settings in the supplied Python code. Minor differences may occur across software versions.

Sources: dashboard_housing(2).py and housing(1).csv supplied in this conversation. No external data were added.